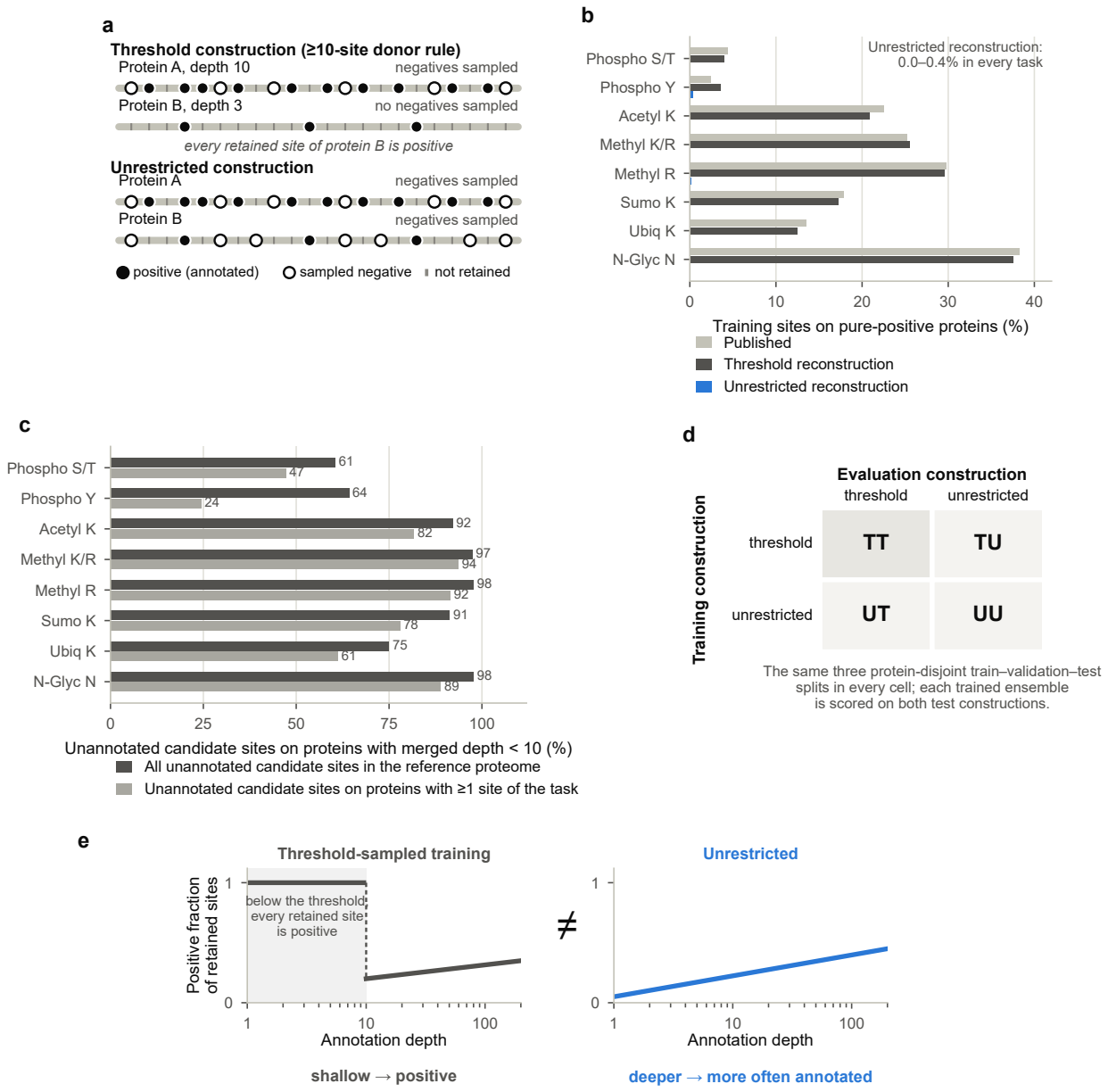

a
Threshold construction (≥10-site donor rule)
Protein A, depth 10
negatives sampled
Protein B, depth 3
no negatives sampled
every retained site of protein B is positive
Unrestricted construction
Protein A
negatives sampled
Protein B
negatives sampled
positive (annotated)
sampled negative
not retained
b
Unrestricted reconstruction:
0.0–0.4% in every task
Phospho S/T
Phospho Y
Acetyl K
Methyl K/R
Methyl R
Sumo K
Ubiq K
N-Glyc N
0
10
20
30
40
Training sites on pure-positive proteins (%)
Published
Threshold reconstruction
Unrestricted reconstruction
c
Phospho S/T 61 47
Phospho Y 64 24
Acetyl K 92 82
Methyl K/R 97 94
Methyl R 98 92
Sumo K 91 78
Ubiq K 75 61
N-Glyc N 98 89
0
25
50
75
100
Unannotated candidate sites on proteins with merged depth < 10 (%)
All unannotated candidate sites in the reference proteome
Unannotated candidate sites on proteins with ≥1 site of the task
d
Evaluation construction
threshold
unrestricted
Training construction
threshold
unrestricted
TT
TU
UT
UU
The same three protein-disjoint train–validation–test splits in every cell; each trained ensemble is scored on both test constructions.
e
Threshold-sampled training
Positive fraction of retained sites
1
0
below the threshold every retained site is positive
1
10
100
Annotation depth
shallow → positive
≠
Unrestricted
1
0
1
10
100
Annotation depth
deeper → more often annotated